

Long term trends: Ovigerous female sea lice

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Mean sea lice dataset (SL_Mean)

```
SL_Mean<-read.table("MeanDataset.csv",  
  header = TRUE, sep = ',', fill = TRUE, dec = ".", na.strings = "NA")
```

```
library(ggplot2)  
library(glmmTMB)  
suppressMessages(library(lmtest))  
library(performance)  
library(gridExtra)  
library(grid)
```

Data adjustment before analysis

The outliers were identified: sites with less than 120 inspections, fish that stayed more than 24 months in the seawater, and outliers (total mobile sea lice mean values higher than 90).

```
# Excluding fish stock transported to sea water in 2023  
SL_Mean1<-SL_Mean[!(SL_Mean$New_Class %in% c("2023 S1/2", "2023 S1")),]  
  
# Sites with less than 120 inspections  
Insp<-aggregate(SL_Mean1$Sp1Date, list(SL_Mean1$SiteName), function(x){length(unique(x))})  
Insp_Sel<-Insp[Insp$x<120,]$Group.1  
  
# Fish more than 24 months in the sea water, group '30' includes 25-30 in seawater  
SL_Mean2<-SL_Mean1[SL_Mean1$TimeSea<30,]  
SL_Mean3<-SL_Mean2[!(SL_Mean2$SiteName %in% Insp_Sel),]  
  
# Outliers  
SL_Pen<-SL_Mean3[SL_Mean3$MeanLepEggs<40,]
```

Comparison of raw and cleaned datasets

```
# Number of counts per fish  
nrow(SL_Mean)
```

```
## [1] 16955
```

```
nrow(SL_Pen)
```

```
## [1] 13494
```

```
# Number of salmon farms  
length(unique(SL_Mean$SiteName))
```

```
## [1] 67
```

```
length(unique(SL_Pen$SiteName))
```

```
## [1] 28
```

```
# Number of mean sea lice values per pen in the dataset  
nrow(SL_Pen)
```

```
## [1] 13494
```

```
mean(SL_Pen$MeanLepTotal);sd(SL_Pen$MeanLepTotal)
```

```
## [1] 3.971273
```

```
## [1] 8.688671
```

```
# Proportion of Total mobile and ovigerous female sea lice  
sum(SL_Pen$MeanLepEggs)/sum(SL_Pen$MeanLepTotal)
```

```
## [1] 0.1810229
```

The format of variables in the dataset (SL_Pen), sea lice values recorded as a mean value per fish per farm, was adjusted.

```
SL_Pen$Year_f<-as.factor(SL_Pen$Year)  
SL_Pen$TimeSea_f<-as.factor(SL_Pen$TimeSea)  
SL_Pen$Year_n<-as.numeric(SL_Pen$Year_f)/10  
SL_Pen$NewMonth<-factor(SL_Pen$NewMonth,levels=c('Jan/Dec','Feb','Mar1','Mar2',  
  'Apr1','Apr2','May1','May2','Jun','Jul','Aug','Sep','Oct','Nov'))
```

A new dataset (SL_Farm), sea lice values recorded as a mean value per fish per farm, was created.

```
SL_Farm<-aggregate(SL_Pen$MeanLepEggs,list(SL_Pen$Year,SL_Pen$Year_f,  
  SL_Pen$SiteName,SL_Pen$NewMonth,SL_Pen$Month,SL_Pen$Class,  
  SL_Pen$TimeSea_f,SL_Pen$Latitude,SL_Pen$Longitude,SL_Pen$SplDate),mean)  
colnames(SL_Farm)<-c('Year','Year_f','SiteName','NewMonth','Month','Class',  
  'TimeSea','Latitude','Longitude','Date','Lep')
```

The format of variables in the dataset (SL_Farm) was adjusted.

```
SL_Farm$TimeSea<-as.factor(SL_Farm$TimeSea)

# Number of mean values per farm in the dataset
nrow(SL_Farm)
```

```
## [1] 7404
```

Year as categorical predictor

The temporal autocorrelation for model at the pen (mean value of sea lice per fish per pen) and at the farm (mean value of sea lice per fish per farm) level were compared.

Pen level

The SL_Pen dataset (mean value per fish per pen) was used in a GLM that included all relevant predictors.

```
m5<-glmmTMB(MeanLepEggs~Year_f+NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
            scale(Longitude)+(1|SiteName),ziformula=~Year_f+NewMonth+
            TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),
            family=ziGamma(link="log"),data=SL_Pen)
```

The temporal autocorrelation was tested.

```
dwtest(m5)
```

```
##
## Durbin-Watson test
##
## data: m5
## DW = 1.9247, p-value = 5.359e-06
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m5)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data: m5
## LM test = 19.183, df = 1, p-value = 1.188e-05
```

Farm level

The SL_Farm dataset (mean value per fish per farm) was used in a GLM that included all relevant predictors.

```
m6<-glmmTMB(Lep~Year_f+NewMonth+TimeSea+Class+scale(Latitude)+
            scale(Longitude)+(1|SiteName),ziformula=~
            Year_f+NewMonth+TimeSea+Class+scale(Latitude)+scale(Longitude),
            family=ziGamma(link="log"),data=SL_Farm)
```

The model was then tested for temporal autocorrelation and the results were compared to the model (m5) at the pen level.

```
dwtest(m6)
```

```
##
## Durbin-Watson test
##
## data: m6
## DW = 1.8567, p-value = 1.362e-11
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m6)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data: m6
## LM test = 38.054, df = 1, p-value = 6.882e-10
```

Lower temporal autocorrelation was detected for the model at the pen level. Based on statistical diagnostics, data availability, and the research question the pen level data (SL_Pen) were used for further analysis.

The model was checked for multicollinearity.

```
check_collinearity(m5)
```

```
## # Check for Multicollinearity
##
## * conditional component:
##
## Low Correlation
##
##      Term  VIF    VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      Year_f 2.24 [2.18, 2.30]   1.01    0.45    [0.43, 0.46]
##      NewMonth 2.50 [2.44, 2.57]   1.04    0.40    [0.39, 0.41]
##      TimeSea_f 2.19 [2.14, 2.25]   1.14    0.46    [0.44, 0.47]
##      Class 1.53 [1.49, 1.56]   1.24    0.65    [0.64, 0.67]
##      CageType 1.05 [1.04, 1.08]   1.03    0.95    [0.93, 0.96]
##      scale(Latitude) 2.68 [2.61, 2.76]   1.64    0.37    [0.36, 0.38]
##      scale(Longitude) 2.69 [2.61, 2.76]   1.64    0.37    [0.36, 0.38]
##
## * zero inflated component:
##
## Low Correlation
##
##      Term  VIF    VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      Year_f 1.45 [1.42, 1.49]   1.01    0.69    [0.67, 0.70]
##      NewMonth 1.84 [1.80, 1.89]   1.02    0.54    [0.53, 0.56]
##      TimeSea_f 1.97 [1.92, 2.02]   1.12    0.51    [0.50, 0.52]
##      scale(Latitude) 2.98 [2.90, 3.07]   1.73    0.34    [0.33, 0.34]
##      scale(Longitude) 2.94 [2.86, 3.02]   1.71    0.34    [0.33, 0.35]
```

The predictors were selected using a backwards step-wise approach.

```
m5.1<-glmmTMB(MeanLepEggs~NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~NewMonth+
  TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m5.2<-glmmTMB(MeanLepEggs~Year_f+TimeSea_f+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year_f+
  TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m5.3<-glmmTMB(MeanLepEggs~Year_f+NewMonth+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year_f+NewMonth+
  Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m5.4<-glmmTMB(MeanLepEggs~Year_f+NewMonth+TimeSea_f+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year_f+NewMonth+
  TimeSea_f+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m5.5<-glmmTMB(MeanLepEggs~Year_f+NewMonth+TimeSea_f+Class+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year_f+NewMonth+
  TimeSea_f+Class+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m5.6<-glmmTMB(MeanLepEggs~Year_f+NewMonth+TimeSea_f+Class+CageType+
  scale(Longitude)+(1|SiteName),ziformula=~Year_f+NewMonth+
  TimeSea_f+Class+CageType+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m5.7<-glmmTMB(MeanLepEggs~Year_f+NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
  (1|SiteName),ziformula=~Year_f+NewMonth+
  TimeSea_f+Class+CageType+scale(Latitude),
  family=ziGamma(link="log"),data=SL_Pen)

AIC(m5,m5.1,m5.2,m5.3,m5.4,m5.5,m5.6,m5.7)
```

```
##      df      AIC
## m5    104 25450.96
## m5.1   44 26469.92
## m5.2   78 27091.12
## m5.3   98 30917.44
## m5.4  102 25491.58
## m5.5  102 25455.42
## m5.6  102 25757.51
## m5.7  102 25567.05
```

```
BIC(m5,m5.1,m5.2,m5.3,m5.4,m5.5,m5.6,m5.7)
```

```
##      df      BIC
## m5    104 26232.00
## m5.1   44 26800.36
## m5.2   78 27676.90
## m5.3   98 31653.42
## m5.4  102 26257.60
## m5.5  102 26221.44
## m5.6  102 26523.53
## m5.7  102 26333.07
```

Best fit was determined for the full model m5. The final model was then tested for overdispersion and the explained variance was determined.

```
# Overdispersion
E2 <- resid(m5, type = "pearson")
N <- nrow(SL_Pen)
p <- length(coef(m5))
sum(E2^2) / (N - p)
```

```
## [1] 1.525394
```

```
# Explained variance
r2_nakagawa(m5)
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.589
##   Marginal R2: 0.530
```

The results of the final model were checked.

```
summary(m5)
```

```
## Family: Gamma ( log )
## Formula:
## MeanLepEggs ~ Year_f + NewMonth + TimeSea_f + Class + CageType +
##   scale(Latitude) + scale(Longitude) + (1 | SiteName)
## Zero inflation:
## ~Year_f + NewMonth + TimeSea_f + Class + CageType + scale(Latitude) +
##   scale(Longitude)
## Data: SL_Pen
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  25451.0  26232.0 -12621.5   25243.0    13390
##
## Random effects:
##
## Conditional model:
##   Groups   Name      Variance Std.Dev.
## SiteName (Intercept) 0.1179   0.3433
## Number of obs: 13494, groups: SiteName, 28
##
## Dispersion estimate for Gamma family (sigma^2): 1.27
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.54899    0.14522  -3.78 0.000157 ***
## Year_f1994    -1.16176    0.13076  -8.88 < 2e-16 ***
## Year_f1995    -1.11963    0.12878  -8.69 < 2e-16 ***
## Year_f1996    -0.54497    0.12374  -4.40 1.06e-05 ***
## Year_f1997    -0.59556    0.12420  -4.80 1.63e-06 ***
## Year_f1998    -0.77283    0.12664  -6.10 1.04e-09 ***
```

```

## Year_f1999      -0.56871    0.12339   -4.61 4.04e-06 ***
## Year_f2000      -1.16060    0.12663   -9.17 < 2e-16 ***
## Year_f2001      -1.31809    0.12690  -10.39 < 2e-16 ***
## Year_f2002      -1.13685    0.12579   -9.04 < 2e-16 ***
## Year_f2003      -0.91791    0.12533   -7.32 2.40e-13 ***
## Year_f2004      -0.64761    0.12826   -5.05 4.44e-07 ***
## Year_f2005      -0.36495    0.12889   -2.83 0.004634 **
## Year_f2006      -0.44298    0.12797   -3.46 0.000537 ***
## Year_f2007      -0.32141    0.12948   -2.48 0.013054 *
## Year_f2008      -0.57032    0.13201   -4.32 1.56e-05 ***
## Year_f2009      -0.74312    0.13343   -5.57 2.56e-08 ***
## Year_f2010      -0.78430    0.14346   -5.47 4.57e-08 ***
## Year_f2011      -1.06023    0.14520   -7.30 2.84e-13 ***
## Year_f2012      -0.85638    0.13662   -6.27 3.65e-10 ***
## Year_f2013      -1.16707    0.14407   -8.10 5.47e-16 ***
## Year_f2014      -0.80320    0.13563   -5.92 3.18e-09 ***
## Year_f2015      -0.77862    0.13445   -5.79 6.99e-09 ***
## Year_f2016      -1.06620    0.14254   -7.48 7.44e-14 ***
## Year_f2017      -1.02474    0.13541   -7.57 3.80e-14 ***
## Year_f2018      -1.52966    0.14487  -10.56 < 2e-16 ***
## Year_f2019      -0.96533    0.13954   -6.92 4.59e-12 ***
## Year_f2020      -1.28435    0.13702   -9.37 < 2e-16 ***
## Year_f2021      -0.84914    0.14379   -5.91 3.52e-09 ***
## Year_f2022      -0.91906    0.14198   -6.47 9.60e-11 ***
## Year_f2023      -2.55723    0.19228  -13.30 < 2e-16 ***
## NewMonthFeb     -0.15675    0.06224   -2.52 0.011786 *
## NewMonthMar1    -0.33418    0.06205   -5.39 7.22e-08 ***
## NewMonthMar2    -0.38483    0.06563   -5.86 4.52e-09 ***
## NewMonthApr1    -0.89127    0.06644  -13.41 < 2e-16 ***
## NewMonthApr2    -0.94232    0.06772  -13.91 < 2e-16 ***
## NewMonthMay1    -1.11894    0.06499  -17.22 < 2e-16 ***
## NewMonthMay2    -0.74477    0.06651  -11.20 < 2e-16 ***
## NewMonthJun     -0.50310    0.06487   -7.76 8.80e-15 ***
## NewMonthJul     -0.38992    0.06575   -5.93 3.02e-09 ***
## NewMonthAug      0.17453    0.06509    2.68 0.007329 **
## NewMonthSep      0.56177    0.06523    8.61 < 2e-16 ***
## NewMonthOct      0.08292    0.06382    1.30 0.193797
## NewMonthNov     -0.06707    0.06132   -1.09 0.274060
## TimeSea_f12      1.11828    0.04946   22.61 < 2e-16 ***
## TimeSea_f18      1.91141    0.04829   39.58 < 2e-16 ***
## TimeSea_f24      2.45368    0.06491   37.80 < 2e-16 ***
## ClassS1/2       -0.24149    0.03723   -6.49 8.74e-11 ***
## CageTypeStd     -0.02881    0.02511   -1.15 0.251329
## scale(Latitude)  0.47017    0.11600    4.05 5.05e-05 ***
## scale(Longitude) -0.25563    0.11258   -2.27 0.023172 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.26712    0.26463    1.01 0.312770
## Year_f1994      0.62464    0.27439    2.28 0.022817 *
## Year_f1995      0.49398    0.27323    1.81 0.070614 .
## Year_f1996     -0.11682    0.27486   -0.43 0.670829

```

```

## Year_f1997      0.53658      0.27129      1.98 0.047943 *
## Year_f1998      1.15646      0.26593      4.35 1.37e-05 ***
## Year_f1999      0.58677      0.26894      2.18 0.029126 *
## Year_f2000      0.91610      0.26655      3.44 0.000588 ***
## Year_f2001      1.33943      0.26434      5.07 4.04e-07 ***
## Year_f2002      0.52282      0.26911      1.94 0.052042 .
## Year_f2003     -0.01208      0.27679     -0.04 0.965198
## Year_f2004      0.37852      0.27696      1.37 0.171722
## Year_f2005      0.30394      0.27768      1.09 0.273712
## Year_f2006      0.06202      0.27817      0.22 0.823560
## Year_f2007      0.30317      0.28106      1.08 0.280732
## Year_f2008      0.48778      0.28053      1.74 0.082069 .
## Year_f2009      0.91529      0.27889      3.28 0.001031 **
## Year_f2010      1.07304      0.28647      3.75 0.000180 ***
## Year_f2011      1.66074      0.27923      5.95 2.72e-09 ***
## Year_f2012      1.33403      0.27668      4.82 1.42e-06 ***
## Year_f2013      1.26745      0.27974      4.53 5.88e-06 ***
## Year_f2014      1.05736      0.27361      3.86 0.000111 ***
## Year_f2015      0.91039      0.27682      3.29 0.001006 **
## Year_f2016      1.67713      0.27743      6.05 1.49e-09 ***
## Year_f2017      0.55243      0.28320      1.95 0.051095 .
## Year_f2018      1.30143      0.28247      4.61 4.08e-06 ***
## Year_f2019      1.14418      0.27767      4.12 3.78e-05 ***
## Year_f2020      0.91203      0.27586      3.31 0.000946 ***
## Year_f2021      1.17711      0.27712      4.25 2.16e-05 ***
## Year_f2022      0.84850      0.28184      3.01 0.002607 **
## Year_f2023      2.45166      0.32465      7.55 4.30e-14 ***
## NewMonthFeb     -0.31288      0.11907     -2.63 0.008595 **
## NewMonthMar1    -0.09472      0.11519     -0.82 0.410920
## NewMonthMar2     0.08866      0.11789      0.75 0.452008
## NewMonthApr1     0.60884      0.12473      4.88 1.05e-06 ***
## NewMonthApr2     0.54015      0.12862      4.20 2.68e-05 ***
## NewMonthMay1     1.16587      0.11281     10.33 < 2e-16 ***
## NewMonthMay2     1.01782      0.11693      8.70 < 2e-16 ***
## NewMonthJun      0.88141      0.11419      7.72 1.17e-14 ***
## NewMonthJul      0.42486      0.11631      3.65 0.000259 ***
## NewMonthAug     -0.45488      0.11995     -3.79 0.000149 ***
## NewMonthSep     -1.03685      0.12514     -8.29 < 2e-16 ***
## NewMonthOct      0.19738      0.11935      1.65 0.098169 .
## NewMonthNov      0.15275      0.12124      1.26 0.207702
## TimeSea_f12     -2.02009      0.06246    -32.34 < 2e-16 ***
## TimeSea_f18     -3.33012      0.06735    -49.44 < 2e-16 ***
## TimeSea_f24     -3.86059      0.12321    -31.33 < 2e-16 ***
## ClassS1/2       0.08505      0.05145      1.65 0.098363 .
## CageTypeStd     -0.12156      0.04553     -2.67 0.007581 **
## scale(Latitude) -0.64076      0.03753    -17.07 < 2e-16 ***
## scale(Longitude) 0.41412      0.03887     10.65 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
car::Anova(m5)
```

```

## Analysis of Deviance Table (Type II Wald chisquare tests)
##

```

```
## Response: MeanLepEggs
##               Chisq Df Pr(>Chisq)
## Year_f        720.0925 30 < 2.2e-16 ***
## NewMonth      1083.8825 13 < 2.2e-16 ***
## TimeSea_f     2018.7648  3 < 2.2e-16 ***
## Class         42.0840  1 8.743e-11 ***
## CageType       1.3159  1  0.25133
## scale(Latitude) 16.4280  1 5.053e-05 ***
## scale(Longitude) 5.1555  1  0.02317 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Visualization of results

The sea lice values for all levels of predictors with the confidential interval were predicted.

```
newdata2 <- data.frame(
  Class=factor(rep(rep(rep(rep(1:2),13),31),14),4),2,levels=1:2,
    labels=levels(as.factor(SL_Pen$Class))),
  Year_f=factor(rep(rep(rep(1:31,each=26),14),4),2,levels=1:31,
    labels=levels(SL_Pen$Year_f)),
  NewMonth=factor(rep(rep(1:14,each=806),4),2,levels=1:14,
    labels=levels(SL_Pen$NewMonth)),
  TimeSea_f=factor(rep(rep(1:4,each=11284),2),levels = 1:4,
    labels =levels(SL_Pen$TimeSea_f)),
  CageType=factor(rep(1:2,each=45136),levels = 1:2,
    labels =levels(as.factor(SL_Pen$CageType))),
  Latitude=mean(SL_Pen$Latitude),
  Longitude=mean(SL_Pen$Longitude))

newdata2 <- cbind(newdata2, predict(m5, newdata2, type = "link", se.fit=TRUE,re.form=NA))

newdata2 <- within(newdata2, {
  Abun<- exp(fit)
  LL <- exp(fit - 1.96 * se.fit)
  UL <- exp(fit + 1.96 * se.fit)
})
```

Mean values of sea lice year were calculated from predicted sea lice counts.

```
A<-aggregate(newdata2$Abun,list(newdata2$Year_f),mean)
B<-aggregate(newdata2$LL,list(newdata2$Year_f),mean)
C<-aggregate(newdata2$UL,list(newdata2$Year_f),mean)
colnames(A)<-c('Year','Abun')
colnames(B)<-c('Year','LL')
colnames(C)<-c('Year','UL')
AA<-merge(A,B, by="Year")
BB<-merge(AA,C, by="Year")
```

The final data were visualized together with the raw data for every year at the level of farm

```

BB$Year<-as.numeric(1993:2023)

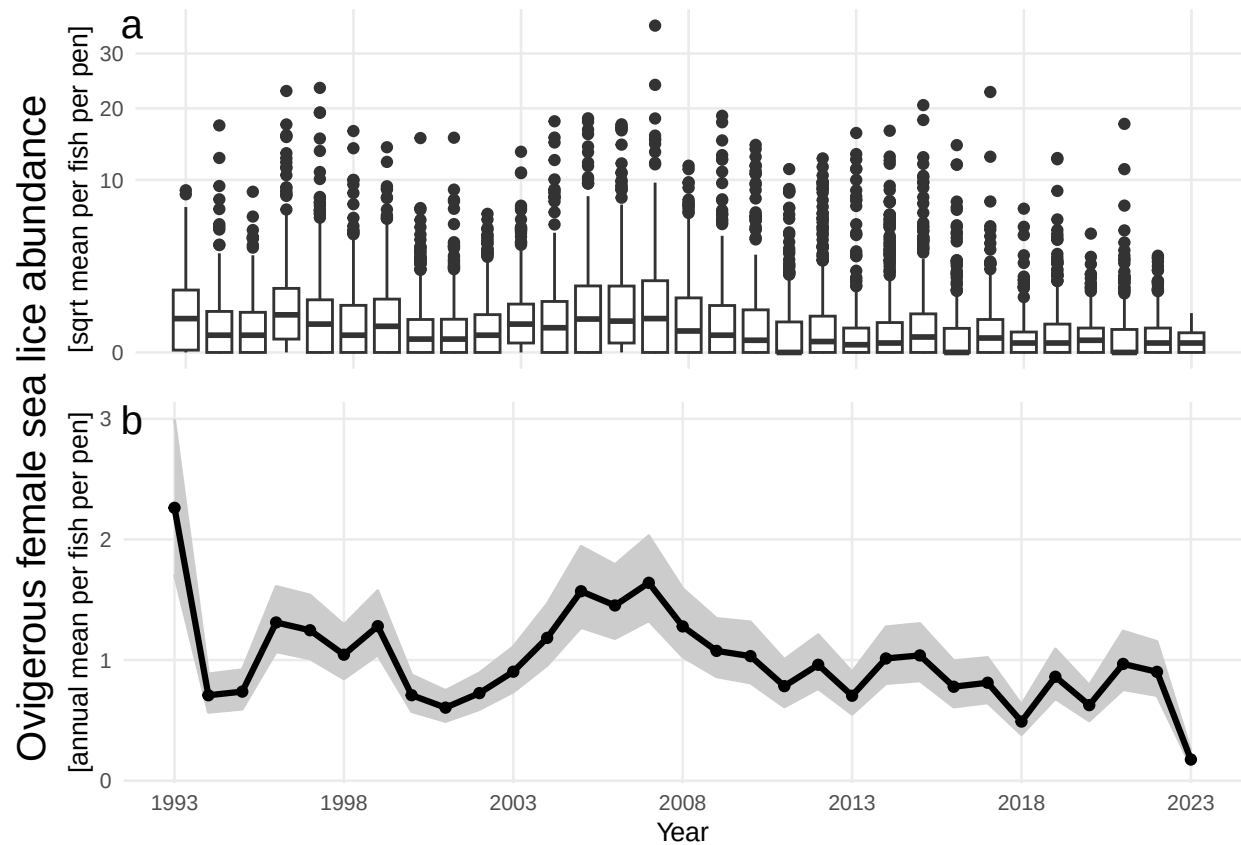
# Predicted sea lice counts
A<-ggplot(BB, aes(Year, Abun)) +
  geom_ribbon(aes(ymin = LL, ymax = UL), colour = 'grey80', fill = "grey80") +
  geom_point() +
  geom_line(linewidth = 1.1) +
  ylab('[annual mean per fish per pen]')+
  theme_minimal(base_size = 10) +
  theme(legend.position = "none",
        panel.grid.minor = element_blank()) +
  scale_x_continuous(name = "Year",limits = c(1992.5, 2023.5),
                     breaks = seq(1993, 2023, 5)) +
  coord_cartesian(xlim = c(1993, 2023))

# Observed sea lice counts
SL_Pen$Year <- as.numeric(as.character(SL_Pen$Year_f))
B<-ggplot(SL_Pen, aes(x = Year, y = MeanLepEggs, group = Year)) +
  geom_boxplot() +
  scale_y_sqrt() +
  ylab('[sqrt mean per fish per pen]')+
  theme_minimal(base_size = 10) +
  theme(legend.position = "none",
        axis.title.x = element_blank(),
        axis.text.x = element_blank(),
        panel.grid.minor = element_blank()) +
  scale_x_continuous(limits = c(1992.5, 2023.5),
                     breaks = seq(1993, 2023, 5)) +
  coord_cartesian(xlim = c(1993, 2023))

grid.arrange(B,A,ncol = 1,
             heights = c(1, 1.2),left = textGrob("Ovigerous female sea lice abundance",
             gp = gpar(fontsize = 15),rot = 90))

# Adding labels A and B
grid.text("a", x = unit(0.10, "npc"), y = unit(.968, "npc"),
          gp = gpar(fontsize = 15, col = "black"))
grid.text("b", x = unit(0.10, "npc"), y = unit(.51, "npc"),
          gp = gpar(fontsize = 15, col = "black"))

```



The predicted values were used to calculate mean values per month to visualize the seasonal changes in sea lice abundance.

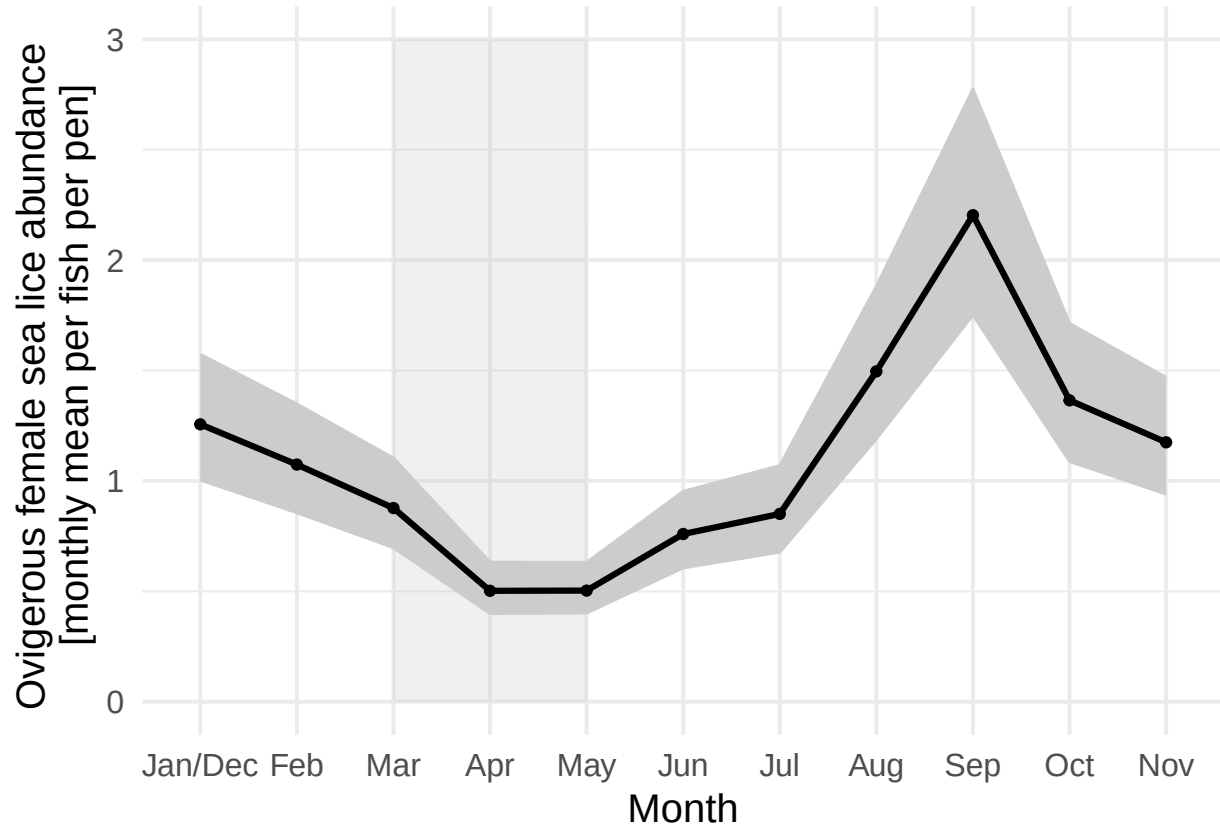
```
newdata2$NewMonth_n<-substr(as.character(newdata2$NewMonth),1,3)
newdata2[newdata2$NewMonth_n=='Jan',]$NewMonth_n<-'Jan/Dec'

A<-aggregate(newdata2$Abun,list(newdata2$NewMonth_n),mean)
B<-aggregate(newdata2$LL,list(newdata2$NewMonth_n),mean)
C<-aggregate(newdata2$UL,list(newdata2$NewMonth_n),mean)
colnames(A)<-c('Month','Abun')
colnames(B)<-c('Month','LL')
colnames(C)<-c('Month','UL')
AA<-merge(A,B, by="Month")
BB<-merge(AA,C, by="Month")

# Common x-axis labels
BB$Month<-factor(BB$Month,levels=c("Jan/Dec","Feb","Mar",
  "Apr","May","Jun","Jul","Aug","Sep","Oct","Nov"))
BLUE <- data.frame(X=c('Mar','Apr','May'), Y = 3)

# Predicted sea lice counts
ggplot(data=BB,aes(Month,Abun,group=1))+
  geom_area(aes(x=X,y=Y),data=BLUE,alpha=0.3,fill="grey80") +
  geom_ribbon(aes(ymin = LL, ymax=UL),colour ='grey80',fill="grey80") +
  geom_point() +
  geom_line(linewidth = 1.1)+
```

```
theme_minimal(base_size =15) +
ylab('Ovigerous female sea lice abundance \n [monthly mean per fish per pen]')+
xlab('Month')+
ylim(0,3)
```



Year as numeric

Pen level

The dataset at the pen level was used to fit a model including all predictors.

```
m7<-glmmTMB(MeanLepEggs~Year+NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),
family=ziGamma(link="log"),data=SL_Pen)
```

The temporal autocorrelation was determined.

```
dwtest(m7)
```

```
##
## Durbin-Watson test
```

```
##
## data:  m7
## DW = 1.9264, p-value = 8.591e-06
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m7)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data:  m7
## LM test = 18.294, df = 1, p-value = 1.893e-05
```

Farm level

The dataset at the farm level was used to fit a model including all predictors.

```
m8<-glmmTMB(Lep~Year+NewMonth+TimeSea+Class+scale(Latitude)+
            scale(Longitude)+(1|SiteName),ziformula=~Year+
            NewMonth+TimeSea+Class+scale(Latitude)+scale(Longitude),
            family=ziGamma(link="log"),data=SL_Farm)
```

The temporal autocorrelation was determined and compared to the model at the pen level.

```
dwtest(m8)
```

```
##
## Durbin-Watson test
##
## data:  m8
## DW = 1.8036, p-value < 2.2e-16
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m8)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data:  m8
## LM test = 71.441, df = 1, p-value < 2.2e-16
```

Lower temporal autocorrelation was detected for the model at the pen level. Based on statistical diagnostics, data availability, and the research question the pen level data (SL_Pen) were used for further analysis.

```
check_collinearity(m7)
```

```
## # Check for Multicollinearity
##
## * conditional component:
##
```

```
## Low Correlation
##
##          Term  VIF    VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##          Year 1.42 [1.39, 1.45]   1.19    0.70    [0.69, 0.72]
##        NewMonth 1.96 [1.92, 2.01]   1.03    0.51    [0.50, 0.52]
##      TimeSea_f 1.96 [1.91, 2.01]   1.12    0.51    [0.50, 0.52]
##          Class 1.40 [1.38, 1.44]   1.19    0.71    [0.70, 0.73]
##        CageType 1.02 [1.01, 1.05]   1.01    0.98    [0.95, 0.99]
##  scale(Latitude) 2.68 [2.61, 2.75]   1.64    0.37    [0.36, 0.38]
##  scale(Longitude) 2.67 [2.60, 2.75]   1.64    0.37    [0.36, 0.38]
##
## * zero inflated component:
##
## Low Correlation
##
##          Term  VIF    VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##          Year 1.16 [1.14, 1.19]   1.08    0.86    [0.84, 0.88]
##        NewMonth 1.76 [1.72, 1.81]   1.02    0.57    [0.55, 0.58]
##      TimeSea_f 1.83 [1.78, 1.87]   1.11    0.55    [0.53, 0.56]
##  scale(Latitude) 2.95 [2.87, 3.03]   1.72    0.34    [0.33, 0.35]
##  scale(Longitude) 2.90 [2.82, 2.98]   1.70    0.34    [0.34, 0.35]
```

The predictors were selected using a backwards step-wise approach.

```
m7.1<-glmmTMB(MeanLepEggs~NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~NewMonth+
  TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m7.2<-glmmTMB(MeanLepEggs~Year+TimeSea_f+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year+
  TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m7.3<-glmmTMB(MeanLepEggs~Year+NewMonth+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
  Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m7.4<-glmmTMB(MeanLepEggs~Year+NewMonth+TimeSea_f+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
  TimeSea_f+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m7.5<-glmmTMB(MeanLepEggs~Year+NewMonth+TimeSea_f+Class+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
  TimeSea_f+Class+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m7.6<-glmmTMB(MeanLepEggs~Year+NewMonth+TimeSea_f+Class+CageType+
  scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
  TimeSea_f+Class+CageType+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m7.7<-glmmTMB(MeanLepEggs~Year+NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
  (1|SiteName),ziformula=~Year+NewMonth+
  TimeSea_f+Class+CageType+scale(Latitude),
  family=ziGamma(link="log"),data=SL_Pen)

AIC(m7,m7.1,m7.2,m7.3,m7.4,m7.5,m7.6,m7.7)
```

```
##      df      AIC
## m7    46 26328.40
## m7.1  44 26469.92
## m7.2  20 27891.70
## m7.3  40 31811.60
## m7.4  44 26366.47
## m7.5  44 26333.51
## m7.6  44 26661.78
## m7.7  44 26447.74
```

```
BIC(m7,m7.1,m7.2,m7.3,m7.4,m7.5,m7.6,m7.7)
```

```
##      df      BIC
## m7    46 26673.86
## m7.1  44 26800.36
## m7.2  20 28041.90
## m7.3  40 32112.00
## m7.4  44 26696.91
## m7.5  44 26663.95
## m7.6  44 26992.22
## m7.7  44 26778.18
```

Best fit was determined for the full model m7. The final model was then tested for overdispersion and the explained variance was determined.

```
# Overdispersion
E2 <- resid(m7, type = "pearson")
N <- nrow(SL_Pen)
p <- length(coef(m7))
sum(E2^2) / (N - p)
```

```
## [1] 1.583126
```

```
# Explained variance
r2_nakagawa(m7)
```

```
## # R2 for Mixed Models
##
## Conditional R2: 0.558
## Marginal R2: 0.498
```

The results of the final model were checked.

```
summary(m7)
```

```
## Family: Gamma ( log )
## Formula:
## MeanLepEggs ~ Year + NewMonth + TimeSea_f + Class + CageType +
## scale(Latitude) + scale(Longitude) + (1 | SiteName)
## Zero inflation:
## ~Year + NewMonth + TimeSea_f + Class + CageType + scale(Latitude) +
```

```

##      scale(Longitude)
## Data: SL_Pen
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 26328.4 26673.9 -13118.2 26236.4    13448
##
## Random effects:
##
## Conditional model:
## Groups   Name      Variance Std.Dev.
## SiteName (Intercept) 0.1158  0.3403
## Number of obs: 13494, groups: SiteName, 28
##
## Dispersion estimate for Gamma family (sigma^2): 1.34
##
## Conditional model:
##      Estimate Std. Error z value Pr(>|z|)
## (Intercept)    23.347140   4.043259   5.77 7.73e-09 ***
## Year           -0.012348   0.002016  -6.12 9.09e-10 ***
## NewMonthFeb     -0.127744   0.063344  -2.02 0.043730 *
## NewMonthMar1    -0.283634   0.063417  -4.47 7.73e-06 ***
## NewMonthMar2    -0.298162   0.066788  -4.46 8.03e-06 ***
## NewMonthApr1    -0.792591   0.067675 -11.71 < 2e-16 ***
## NewMonthApr2    -0.881236   0.068870 -12.80 < 2e-16 ***
## NewMonthMay1    -1.070285   0.066153 -16.18 < 2e-16 ***
## NewMonthMay2    -0.711531   0.067740 -10.50 < 2e-16 ***
## NewMonthJun     -0.465717   0.065878  -7.07 1.56e-12 ***
## NewMonthJul     -0.348663   0.067026  -5.20 1.97e-07 ***
## NewMonthAug      0.226958   0.066248   3.43 0.000613 ***
## NewMonthSep      0.608386   0.066440   9.16 < 2e-16 ***
## NewMonthOct      0.121568   0.064843   1.87 0.060819 .
## NewMonthNov     -0.072062   0.062418  -1.15 0.248290
## TimeSea_f12      1.121533   0.050563  22.18 < 2e-16 ***
## TimeSea_f18      1.928981   0.048671  39.63 < 2e-16 ***
## TimeSea_f24      2.519848   0.065513  38.46 < 2e-16 ***
## ClassS1/2       -0.236981   0.037254  -6.36 2.00e-10 ***
## CageTypeStd     -0.007218   0.025404  -0.28 0.776323
## scale(Latitude)  0.501812   0.115000   4.36 1.28e-05 ***
## scale(Longitude) -0.281435   0.111499  -2.52 0.011600 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##      Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -56.570940   5.531842 -10.23 < 2e-16 ***
## Year           0.028706   0.002759  10.40 < 2e-16 ***
## NewMonthFeb    -0.307823   0.116860  -2.63 0.008436 **
## NewMonthMar1   -0.121003   0.113121  -1.07 0.284763
## NewMonthMar2    0.051590   0.115617   0.45 0.655441
## NewMonthApr1    0.580086   0.122331   4.74 2.12e-06 ***
## NewMonthApr2    0.511117   0.126362   4.04 5.23e-05 ***
## NewMonthMay1    1.094522   0.110434   9.91 < 2e-16 ***
## NewMonthMay2    0.953009   0.114578   8.32 < 2e-16 ***
## NewMonthJun     0.819669   0.111867   7.33 2.35e-13 ***

```

```
## NewMonthJul      0.387915  0.114141   3.40 0.000677 ***
## NewMonthAug     -0.461940  0.117640  -3.93 8.61e-05 ***
## NewMonthSep     -1.019686  0.122799  -8.30 < 2e-16 ***
## NewMonthOct      0.151047  0.117043   1.29 0.196868
## NewMonthNov      0.131922  0.119184   1.11 0.268347
## TimeSea_f12     -1.932473  0.060696 -31.84 < 2e-16 ***
## TimeSea_f18     -3.214185  0.064890 -49.53 < 2e-16 ***
## TimeSea_f24     -3.692708  0.119114 -31.00 < 2e-16 ***
## ClassS1/2       0.066666  0.049692   1.34 0.179730
## CageTypeStd     -0.133415  0.044463  -3.00 0.002694 **
## scale(Latitude) -0.650175  0.036563 -17.78 < 2e-16 ***
## scale(Longitude) 0.406938  0.037836  10.76 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
car::Anova(m7)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: MeanLepEggs
##              Chisq Df Pr(>Chisq)
## Year          37.5099  1  9.095e-10 ***
## NewMonth     1026.6232 13 < 2.2e-16 ***
## TimeSea_f    2096.7751  3 < 2.2e-16 ***
## Class         40.4649  1  2.002e-10 ***
## CageType       0.0807  1    0.7763
## scale(Latitude) 19.0408  1  1.280e-05 ***
## scale(Longitude) 6.3711  1    0.0116 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```